The Reversible Ledger

5 instances · Download all inputs (.zip)

The Reversible Ledger

Deep in the vaults of the **Bureau of Symmetric Accounting**, every entry must be audited against its own reflection. An auditor takes a positive integer `n`, writes it backwards to obtain `reverse(n)`, and computes the **reflected sum** `n + reverse(n)`. The entry passes the audit iff **every digit of that sum is odd**. Such a number is then called **reversible**

Two rules of the Bureau make the reflection well defined:

- Neither `n` nor `reverse(n)` may have a leading zero. In particular, `n` must not end in `0` (otherwise its reflection would start with a `0`).
- The reflected sum is read digit by digit; **all** of its digits must be odd (`1`, `3`, `5`, `7`, `9`).

For example, `36` reflects to `63`, and `36 + 63 = 99`: both digits are odd, so `36` is reversible. Likewise `409 + 904 = 1313`, all odd, so `409` passes. But `13 + 31 = 44` has even digits and fails, and `10` ends in `0`, so it is disqualified outright, even though `10 + 01 = 11`.

The head auditor wants a tally. Given a size limit, count how many entries pass.

Goal

Given an integer `K`, determine how many **reversible** positive integers are strictly smaller than `10^K` (that is, the reversible numbers with **at most `K` digits**).

Input Format

A single line containing one integer `K`.

Output Format

A single integer: the number of reversible positive integers below `10^K`.

Constraints

Instances are guaranteed to satisfy the following constraints:

- `1 ≤ K ≤ 18`
- The answer fits in a 64-bit signed integer.

Example

Instance in input:

```
2
```

Solution in output:

```
20
```

Below `10^2` the reversible numbers are exactly the two-digit numbers `10a + b` (with `a, b` from `1` to `9`) whose digit sum `a + b` is odd and at most `9`: the reflected sum is `11·(a + b)`, i.e. the two equal digits `(a + b)(a + b)`, which are odd precisely when `a + b` is odd. Counting the ordered pairs with `a + b ∈ {3, 5, 7, 9}` gives `2 + 4 + 6 + 8 = 20`. No single-digit number is reversible, since `n + reverse(n) = 2n` always ends in an even digit.

Instances

Submit all instances at once

Upload one .zip containing a file per instance, each named after its instance number (e.g. `instance.1.out`, `2.txt`).

Browse... No file selected.

Submit all

Your code

Upload a .zip of the code your team used to generate the answers for this problem. Re-uploading replaces the previous file.

Current upload: download your code

Browse... No file selected.

Upload code

Instance 1 SOLVED ✓

▸ Show input

Paste your answer

Paste output here, or upload a file below

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Submit instance 1

Instance 2 SOLVED ✓

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Instance 3 SOLVED ✓

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Instance 4

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Instance 5

▸ Show input

Paste your answer

Paste output here, or upload a file below

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Submit instance 5